Type MS · Describe DNA's double helix, base pairing and replication

Answer 1 of 10 · Question 3 part c · 4 marks

2024 JUN · Paper 01 · find part c in this mark scheme extract

3 (a) (i)		nucleus;		(1)
	(ii)	allele;		(1)
	(iii)	too many fingers/toes;		(1)
(b) (i)		heterozygous/heterozygote/carrier;		(1)
	(ii)	Y;		(1)
(c)		Any four from: • <u>double</u> helix/two strands; • consists of nucleotides; • four bases; • each strand held together by hydrogen bonds/bases linked by hydrogen bonds; • complementary base pairing/A-T, G-C;		(4)

Total Question 3 = 9 marks

Type MS · Describe DNA's double helix, base pairing and replication

Answer 2 of 10 · Question 1 part a.iii · 2 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

1 (a) (i)	**D** sugars A/B/C are incorrect as amino acids/bases/protein do not form part of backbone		1
(ii)	**B** bases A is incorrect as the structures labelled Y are bases C is incorrect as proteins are not incorporated into DNA structure D is incorrect as sugar is found in the backbone		1
(iii)	Any two from the following: • hydrogen bonds; • between complementary bases;	Allow description of complementary base pairing	2
(iv)	nucleus;		1
(b) (i)	Any two from the following: • RNA contains ribose sugar and DNA contains deoxyribose; • RNA is single-stranded and DNA is double stranded; • RNA contains uracil and DNA contains thymine;		2
(ii)	**D** mRNA and tRNA A/B/C are incorrect as pRNA/dRNA/dRNA are not types of RNA		1
(c)	**R** phosphate; **S** sugar/deoxyribose; **T** base;	Allow a named base	3
			Total 11

Type MS · Describe DNA's double helix, base pairing and replication

Answer 2 of 10 · Question 1 part a.iii · 2 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe DNA's double helix, base pairing and replication

Answer 3 of 10 · Question 1 part b.ii · 3 marks

2023 JUN · Paper 02 (regional R) · find part b.ii in this mark scheme extract

Number				
1 (a) (i)	<table><tr><th rowspan="2">cell structure</th><th colspan="3">Function</th></tr><tr><th>makes proteins</th><th>contains DNA</th><th>allows substances in and out of cells</th></tr><tr><td>nucleus</td><td></td><td>✓</td><td></td></tr><tr><td>ribosomes</td><td>✓</td><td></td><td></td></tr><tr><td>cell membrane</td><td></td><td></td><td>✓</td></tr></table>	One mark for each correct row	3
(ii)	cell membrane/cell wall/cytoplasm/plasmid/DNA/ribosomes/flagellum;		1
(b) (i)	Part W phosphate (group); Part X sugar/ deoxyribose sugar;		2
(ii)	• complementary base pairing; • A-T, G-C; • hydrogen bonds between bases;		3
(c)	153 ÷ 3; 51;	Full marks for correct final answer Ecf allow correct final answer to calculation for 1 mark	2

Total for Question 1 = 11 Marks

Type MS · Describe DNA's double helix, base pairing and replication

Answer 4 of 10 · Question 2 part b.i · 1 mark

2025 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

2 (a)	(i)	X = (pentose) sugar/deoxyribose; Y = phosphate;	reject ribose reject phosphorus	2
	(ii)	DNA/deoxyribonucleic acid;		1
	(iii)	• double/two stranded; • thymine present; • only found in DNA (not RNA);		3
	(iv)	nucleus;		1
(b)	(i)	hydrogen bonds;		1
	(ii)	• connect/hold/join, two strands/bases, together; • between complementary bases (pairs)/nucleotides; • adenine thymine/guanine cytosine;	ignore just A T G C	3
				11 marks

Type MS · Describe DNA's double helix, base pairing and replication

Answer 4 of 10 · Question 2 part b.i · 1 mark

2025 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe DNA's double helix, base pairing and replication

Answer 5 of 10 · Question 2 part b.ii · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

2 (a) (i)		X = (pentose) sugar/deoxyribose; Y = phosphate;	reject ribose reject phosphorus	2
	(ii)	DNA/deoxyribonucleic acid;		1
	(iii)	• double/two stranded; • thymine present; • only found in DNA (not RNA);		3
	(iv)	nucleus;		1
(b) (i)		hydrogen bonds;		1
	(ii)	• connect/hold/join, two strands/bases, together; • between complementary bases (pairs)/nucleotides; • adenine thymine/guanine cytosine;	ignore just A T G C	3
				11 marks

Type MS · Describe DNA's double helix, base pairing and replication

Answer 5 of 10 · Question 2 part b.ii · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe DNA's double helix, base pairing and replication

Answer 6 of 10 · Question 8 part c.i · 1 mark

2020 NOV · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

8 (a) (i)		chromosomes (1)		1
	(ii)	 (1)  (1)		2
(b) (i)		complementary bases pair up 3 DNA double helix unwinds 1 strands separate 2 two DNA strands form 4		1
(c) (i)		thymine/T (1)	R thiamine/thyamine	1
	(ii)	guanine/G/cytosine/C (1)		1
(c)		any four from • allows formation of gametes (1) • haploid (1) • allows variation to occur (1) • which allows species to evolve (1) • allows diploid number to be maintained (1) • at fertilisation (1)		4

Total 10 marks

Type MS · Describe DNA's double helix, base pairing and replication

Answer 6 of 10 · Question 8 part c.i · 1 mark

2020 NOV · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

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Type MS · Describe DNA's double helix, base pairing and replication

Answer 7 of 10 · Question 8 part c.ii · 1 mark

2020 NOV · Paper 01 · extract 1 of 2 · find part c.ii in this mark scheme extract

8 (a) (i)		chromosomes (1)		1
	(ii)	 (1)  (1)		2
(b) (i)		complementary bases pair up 3 DNA double helix unwinds 1 strands separate 2 two DNA strands form 4		1
(c) (i)		thymine/T (1)	R thiamine/thyamine	1
	(ii)	guanine/G/cytosine/C (1)		1
(c)		any four from • allows formation of gametes (1) • haploid (1) • allows variation to occur (1) • which allows species to evolve (1) • allows diploid number to be maintained (1) • at fertilisation (1)		4

Total 10 marks

Type MS · Describe DNA's double helix, base pairing and replication

Answer 7 of 10 · Question 8 part c.ii · 1 mark

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Type MS · Describe DNA's double helix, base pairing and replication

Answer 8 of 10 · Question 6 part b · 2 marks

2020 NOV · Paper 01 (regional R) · find part b in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Describe DNA's double helix, base pairing and replication

Answer 9 of 10 · Question 8 part b · 1 mark

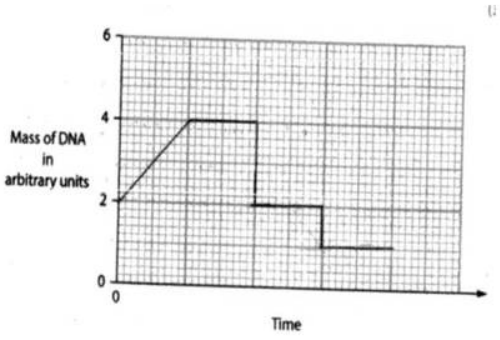
2020 NOV · Paper 01 · find part b in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Describe DNA's double helix, base pairing and replication

Answer 10 of 10 · Question 4 part b.i · 4 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Number					
4	(a)	(i)	**A; (chromosomes)** B/C/D do not contain DNA		1
		(ii)	RNA/mRNA/tRNA;		1
		(iii)	<table><tr><th>Differences</th><th>DNA</th><th>Named nucleic acid</th></tr><tr><td>1</td><td>double stranded</td><td>single stranded</td></tr><tr><td>2</td><td>contains deoxyribose</td><td>contains ribose</td></tr><tr><td>3</td><td>contains thymine</td><td>contains uracil</td></tr></table>		3
	(b)	(i)	• replication; • formation of a new strand (of DNA); • by complementary base pairing; • double the original mass of DNA;		4
		(ii)	• cell dividing; • cytokinesis;		2
		(iii)	• line starting at two; • increases to four; • ends at one;  		3

Total 14 marks

Type MS · Describe DNA's double helix, base pairing and replication

Answer 10 of 10 · Question 4 part b.i · 4 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks